

Appendix 17.A Two-port networks and S-parameters

Many of the components that are used in wireless transceivers can be described as *two-ports*, i.e., having an input port and an output port. While the voltages and currents at those ports characterize the circuit, it is more common to describe them by incident and reflected *complex* waves, together with a *characteristic impedance* Z_0 , commonly chosen as real-valued, and in particular 50Ω . Shown in Fig. 17.24 are the incident waves a_i and outgoing waves b_i at the two ports. They have the dimension "square root of power", and are related to the voltage and current at the ports as

$$a_i = \frac{V_i + Z_0 I_i}{2\sqrt{Z_0}}, \quad b_i = \frac{V_i - Z_0 I_i}{2\sqrt{Z_0}}. \quad (17.50)$$

The relation between these waves is described by the scattering parameters, or in short *S-parameters*

$$b_1 = S_{11}a_1 + S_{12}a_2 \quad (17.51)$$

$$b_2 = S_{21}a_1 + S_{22}a_2.$$

These four S-parameters can be written into a 2×2 matrix

$$S = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}. \quad (17.52)$$

For a reciprocal two-port, the S matrix is symmetric, and for a lossless two-port, the S-matrix is unitary. Also note that the S-parameters are defined with respect to a particular reference plane, and that changing the reference plane results in a phase shift of the S-parameters.

From (17.51) it follows immediately that $|S_{11}|^2$ and $|S_{22}|^2$ describe the fraction of power that gets reflected back at ports 1 and 2, respectively *when source and load impedance are Z_0* . The reflection coefficient at the input when the load impedance across the output is an arbitrary value Z_L is

$$\Gamma_{in} = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L} \quad \text{where} \quad \Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}. \quad (17.53)$$

The reflection coefficient at the output for arbitrary source impedance Z_s is completely analogous

$$\Gamma_{out} = S_{22} + \frac{S_{12}S_{21}\Gamma_s}{1 - S_{11}\Gamma_s} \quad \text{where} \quad \Gamma_s = \frac{Z_s - Z_0}{Z_s + Z_0}. \quad (17.54)$$

These expressions are also important since they allow to test stability: $|\Gamma_{in}| < 1$ and $|\Gamma_{out}| < 1$ must be fulfilled to avoid positive feedback and thus unbounded oscillations. If stability can be achieved for any load (or source) impedance, then the stability is called unconditional; otherwise it is conditional. It is also important that stability has to be ensured over all frequencies of interest; an oscillation at even one frequency can affect all other frequencies. The reflection coefficient is related to the *Voltage Standing Wave Ratio*, *VSWR*, which is defined as

$$VSWR = \frac{1+|\Gamma|}{1-|\Gamma|}, \quad (17.55)$$

The *transducer power gain*, which is the power delivered to a load impedance Z_L divided by the power that would be delivered to a conjugate-matched load, is

$$G_T = \frac{(1-|\Gamma_s|^2)|S_{21}|^2(1-|\Gamma_L|^2)}{|1-\Gamma_s\Gamma_{in}|^2|(1-S_{22}\Gamma_L)|^2} \quad (17.56)$$

$$= \frac{(1-|\Gamma_s|^2)|S_{21}|^2(1-|\Gamma_L|^2)}{|(1-S_{11}\Gamma_s)(1-S_{22}\Gamma_L)-S_{12}S_{21}\Gamma_s\Gamma_L|^2}. \quad (17.57)$$

Besides this transducer power gain, we can also define the *power gain* as the ratio of power delivered to the load impedance divided by the power delivered to the input of the network

$$G_p = \frac{|S_{21}|^2(1-|\Gamma_L|^2)}{(1-|\Gamma_{in}|^2)|1-S_{22}\Gamma_L|^2} \quad (17.58)$$

where Γ_{in} is the reflection coefficient seen "looking into" the network at port 1 (with the output connected to the actual load). We can also define the *available gain*, which is the power available from the two-port network, divided by the power available from the source.

S-parameters are usually measured with *Vector Network Analyzers* (VNAs). A VNA is generally first calibrated (by letting it measure an open circuit, a short, and a through connection), and then measures all the 4 S parameters of a two-port. VNAs are among the standard pieces of equipment in any RF laboratory. Lower-cost devices can measure up to about 6-10 GHz, and thus cover most cellular bands; millimeter-wave capable VNAs (typically ranging up to 40, or 67 GHz, and whose range can be extended with suitable add-ones to several hundred GHz) are commercially available but rather expensive.

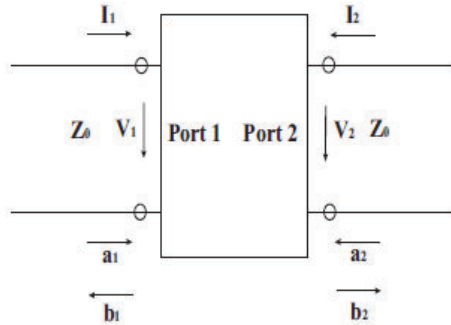


Figure 17.24. Two port network. Shown are the voltages, currents, and incident and outgoing waves.

Appendix 17.B Matching

Overview

In this appendix we discuss in more detail how to design matching circuitry and the performance that can be achieved with it.

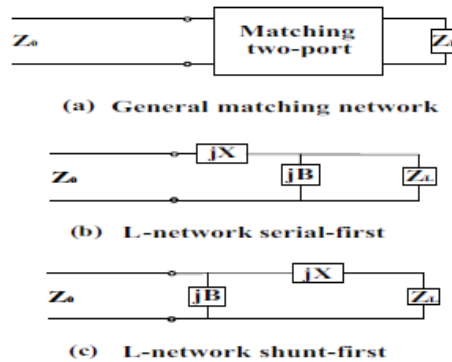


Figure 17.25. (a) General matching network. (b). L-network serial-first (c). L-network shunt-first.

Matching can be used for the following purposes:

- *Elimination of reflections*: A jump in the impedance that occurs at the interface between two components leads to reflections. Reflections are undesirable in microwave circuits, because they may damage sensitive components, and the echoes of a signal might lead to intersymbol interference (similar to the ISI created by the propagation channel, compare Chapter 2). If the input of the matching two-port is connected to a source with impedance Z_s , reflections are avoided if the matching network is designed such that the impedance "looking into" the matching network (which is terminated by the load) is also Z_s^* .
- *Maximization of transferred power*: the relationship between input and output impedance, and a possible network in between, determines the amount of power that can be transferred to a load. To maximize this transferred power, the impedance looking into the matching network should be the complex conjugate of the source impedance, Z_s^* .
- *Minimization of noise*: the noise generated in an amplifier depends on the signal source impedance. It might thus be desirable to transform, by means of a two-port network, the actual signal source impedance to a value that equals this optimum.
- *Decorrelation of antennas*: the matching network at the connectors of an antenna array can impact the correlation of the signals at these elements. Since decorrelated signals are generally desirable to enhance diversity, and increase capacity of spatial multiplexing, such decorrelation matching has drawn some attention in recent years.

Note that the above criteria are different from each other, and a system designer either has to pick the most important criterion, or find some compromise between the different targets.

Generally, implementations of matching networks are distinguished by

- *Discrete-component versus distributed-component*: at low frequencies, up to a few hundred MHz, matching can be achieved by two-ports consisting of discrete components such as inductor and capacitor. For higher frequencies, matching networks made from distributed components such as transmission lines are preferred.

- *Narrowband versus wideband:* depending on the system bandwidth, it might be sufficient to achieve matching at only a single frequency, or it must work over a wider bandwidth. The bandwidth of a matching circuit can be described either by the 3 dB bandwidth B_Q , or by the Q -factor (quality factor, compare also Section 8.2), which is usually defined as the ratio of the energy stored in the circuit and escaped energy per cycle of the oscillation. The Q -factor is approximately the inverse relative bandwidth

$$Q = \frac{\omega_c}{2\pi B_Q} . \quad (17.59)$$

- *Lossless versus lossy:* in most circumstances, a matching network is desired to be lossless, and thus created from purely reactive components such as capacitors, inductances, and transmission lines. However, in some circumstances, lossy matching using resistive elements might be useful, because it provides matching with a wider bandwidth.

Narrowband matching with discrete components

The simplest lossless two-port for matching consists of two reactive components, which can be arranged in the structure of either serial-first, or shunt-first, see Fig. 17.25b,c. We generally write $Z = R + jX$ and $1/Z = G + jB$. It is easy to write analytic equations for the matching, though the solutions can sometimes become cumbersome. For example, for the serial-first structure (Fig. 17.26b), the input impedance seen at the reference plane should be equal to the transmission line impedance, which is assumed to be at the reference impedance Z_0

$$jX_{\text{ser}} + \frac{1}{jB_{\text{shu}} + \frac{1}{R_L + jX_L}} = Z_0 . \quad (17.60)$$

From this equation, the suitable values for B_{ser} and X_{shu} can be computed from two equations, which set equal the real and the imaginary parts on the left and right side of Eq. (17.60); note that often two solutions are possible, and it may depend on other criteria (e.g., values of inductors or capacitances that can be easily manufactured) which ones to pick. Similar equations can be derived for the shunt-first structure. Also, note that not all structures of Fig. 17.25b,c can be applied to all matching problems.

Wideband matching with discrete components

It is clear from the above description that the matching is performed at only one frequency, since X is either ωL_{coil} for an inductor (with L_{coil} being the inductance of a coil), or $-1/(\omega C_{\text{cap}})$ for a capacitor. Thus, a change of frequency changes X , and thus the matching. The bandwidth over which the matching approximately works is determined by the Q , which in turn depends on the values of the components that are being used. For example, the serial-first matching circuit provides $Q = X_{\text{ser}} / R_L$.

In many situations, a more wideband matching is desired. This can be achieved by using a more elaborate matching structure that, in essence, executes the matching in multiple steps. If, for example, a matching from a low-impedance load to a higher Z_0 is desired, then we can first transform to an intermediate impedance (often chosen to be the geometric mean of R_L and Z_0), and from there to Z_0 . Such a two-stage matching structure is simply a cascade of the simple structure from Fig. 17.25b (or Fig. 17.25c, as may be the case), and provides a larger bandwidth than a single stage.

In some situations it is also desirable to make the matching more narrowband, to obtain an implicit filtering of the signal. This can be achieved by using combinations of serial-first and shunt-first structures. A shunt-first

followed by a serial-first, also known as Π -circuit, while serial-first followed by shunt-first is called a T -circuit. Also for these, analytical solutions can be obtained, see [Pozar 2005].

Besides the analytic solutions, there are also helpful graphic solutions that are based on the *Smith-Chart*, a graph that shows the complex reflection coefficient (usually within the unit circle), as well as curves that characterize constant R , and constant X (this form is called the normalized impedance graph), or curves for constant G and constant B (the normalized admittance graph). Perfect matching corresponds to location of the reflection coefficient at the center of the circle. The Smith chart used to be a standard tool for development of matching circuits, but has been mostly superseded by software programs; it is now mainly used to display the output of such programs, demonstrating how RF parameters change over frequency.

As already discussed in Chapter 8, there is a limit to the quality of the matching over a large bandwidth. The Bode-Fano criterion provides a fundamental bound. While previously we only discussed the bounds that are relevant for the equivalent circuits of antennas, there are actually more general bounds for a number of RLC circuits, see Fig. 17.26.

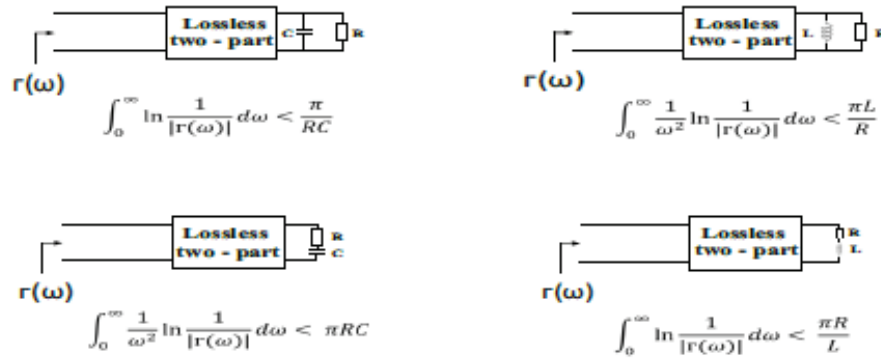


Figure 17.26. Bode-Fano limits for matching by a lossless network.

Matching with distributed components

At higher frequencies, concentrated components show severe nonidealities - for example, the capacitances between the turns of a coil become non-negligible. Furthermore, the length of the transmission lines (measured in units of wavelength) between the different components may become non-negligible. Turning a bug into a feature, it then becomes possible to perform matching by means of (suitable designed) transmission lines.

To design matching networks from transmission lines, first analyze the reflection coefficient that is caused by a jump in impedance between the characteristic impedance of the line and the load

$$\Gamma = \frac{Z_L - Z_{TL}}{Z_L + Z_{TL}}. \quad (17.61)$$

Furthermore, a transmission line provides a phase shift $2(2\pi l_{TL} / \lambda)$, where l_{TL} is the geometric length of the transmission line and λ is the wavelength.¹ The factor 2 at the beginning is caused by the fact that a reflected wave travels forward and back over the length of the line. The input impedance is thus

¹ Remember that the wavelength on a microstrip line is different from the wavelength in vacuum, and depends on the dielectric constant of the substrate on which the line is placed, see Sec. 8.3.3.

$$Z_{in} = Z_{TL} \frac{1 + \Gamma \exp(-2j(2\pi l_{TL} / \lambda))}{1 - \Gamma \exp(-2j(2\pi l_{TL} / \lambda))}. \quad (17.62)$$

Both changes of the characteristic impedance, and the length of the microstrip line, can thus change the input impedance.

An especially simple way to provide matching with the help of a transmission line uses quarter-wave length lines with impedance

$$Z_{TL} = \sqrt{Z_{in} Z_L}. \quad (17.63)$$

This provides the desired matching if the product $Z_{in} Z_L$ is real-valued.

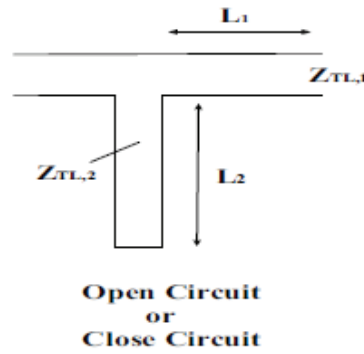


Figure 17.27. Impedance matching with a transmission line and a shunt stub.

A more general way of matching with transmission lines is a combination of a serial transmission line (as discussed above) with a stub that leads away (usually at right angle) from the connection point, see Fig. 17.27 for the geometry of the layout. In essence, the transmission line with length l_1 serves to match the real part, while the stub serves to compensate the imaginary part of the impedance. To provide more detail: we first pick $Z_{TL,1}$ and $Z_{TL,2}$ (often the same value) that result in easy-to-manufacture transmission lines. Then we compute l_1 as a value that matches the input admittance to the source admittance, i.e., makes $\text{Re}\{1/Z_{in}\} = \text{Re}\{1/Z_{source}\}$ according to Eq. (17.62). We then decide whether the stub should have an open or short circuit at the end. Let Y_1 denote the imaginary part of the input admittance of the two-port plus-load at the location of the stub. The sum of the imaginary part of the admittance of the stub, B_{stub} and Y_1 should be either matched to the imaginary part of the source admittance (for zero reflections) or matched to the negative value of the source admittance (for conjugate matching).

The above matching principles are only some very simple (though practically useful) examples; other variants exist. For example, the transformation by means of a simple transmission line can be replaced by a series of transmission lines with progressively changing cross sections, which in the limit become a tapered transmission line. Also, the single stub can be replaced by two or more stubs. All of these measures improve the flexibility of the design. However, it is worth keeping in mind that all of these matching circuits are subject to the Bode-Fano limits.